

LABAD: Behavioral Bifurcation Detection for Insider Threats

An LSTM-Autoencoder UEBA Pipeline with Online Changepoint Onset Localization

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June 21, 2026

Abstract

Most user-and-entity behavior analytics (UEBA) systems answer a single question – *is this user anomalous?* – and report an AUC. The question a security analyst actually asks is *when did the compromise begin?*, because that scopes the investigation. LABAD is an insider-threat detection pipeline on the CERT r4.2 dataset that answers both. An LSTM autoencoder trained only on benign behavior produces per-user anomaly scores; a Neyman–Pearson rule calibrates the operating point to a fixed false-positive budget; and—the central contribution—*Bayesian Online Changepoint Detection (BOCPD)* localizes the onset of malicious activity by treating each user’s behavior as a dynamical system and detecting the moment its statistical regime changes: a behavioral *bifurcation*. The system reaches AUC 0.87 and 100% precision@10 for detection, and localizes the onset of abrupt insiders (CERT scenario 1) to within ± 7 days in $\approx 80\%$ of cases, a median of 3 days, typically 1–2 days *before* the labeled onset. We additionally report rigorous negative and positive results. As causal *detectors*, CUSUM and critical slowing down fail, root-caused to the signal (heavy autocorrelation; scalar dilution of a $151\times$ single-feature spike into a $1.03\times$ non-event); we further show that gradual insiders (scenario 2) are *unlocalizable at the label itself*, because the discriminating behaviors are more active before the labeled onset than after. As an *early-warning* layer, by contrast, critical slowing down yields a statistically significant $2.28\times$ enrichment of pre-onset signatures (24/30 abrupt insiders, $p < 10^{-3}$), confirming the dynamical-systems prediction. The pipeline is containerized (Docker, CPU-only 2.93 GB image) and experiment-tracked (MLflow).

1 Introduction

Insider threats are among the costliest and hardest-to-detect security risks: the actor already has legitimate access, so signature- and rule-based controls do not fire. UEBA reframes the problem as anomaly detection over behavior. The dominant academic approach—an LSTM or autoencoder over the CERT insider-threat dataset—is by now well-trodden, and typically stops at a discrimination metric.

LABAD’s thesis is that the operationally valuable output is not just a risk score but an *onset timestamp*. We formalize “when did the compromise begin” as changepoint detection on each user’s anomaly-score time series, grounded in a dynamical-systems view: a benign user occupies a stable behavioral regime, and going rogue is a *bifurcation*—a sharp qualitative transition to a new regime. BOCPD detects exactly this, online.

Contributions.

1. An end-to-end UEBA pipeline: feature engineering $\rightarrow$ LSTM-autoencoder anomaly scoring $\rightarrow$ Neyman–Pearson calibration $\rightarrow$ BOCPD onset detection $\rightarrow$ RAG-grounded LLM threat reports, containerized and tracked.
2. A from-scratch BOCPD implementation with a corrected changepoint readout, and an evaluation showing onset localization to a median of 3 days on abrupt insiders.
3. Rigorous *negative* results: CUSUM and critical slowing down do not improve on BOCPD, root-caused to signal autocorrelation and scalar-score dilution rather than detector choice.
4. A per-feature ablation that recovers the diluted signal and adds feature-level explainability, and a demonstration that gradual-insider onset is unlocalizable at the CERT label itself.
5. A statistically significant ($p < 10^{-3}$) critical-slowing-down early-warning signal preceding abrupt onsets, validating the dynamical-systems framing as predictive, not merely descriptive.

2 Data and Feature Engineering

We use **CERT r4.2**, a synthetic but richly simulated insider-threat corpus (≈ 16 GB of raw logs across logon, device/USB, file, email, and HTTP activity), with ground-truth incidents in `insiders.csv`. After filtering to release 4.2 there are 70 labeled malicious users across three attack scenarios.

Raw events are aggregated into **per-user, per-day** feature vectors of 18 behavioral dimensions: logon counts and after-hours/weekend logons, first/last activity hour, USB connect counts (and after-hours USB), file access and after-hours file access, executable access, unique files, email counts (external, with-attachment, average size), HTTP counts, job-site visits, and cloud-storage visits. A day is the smallest unit at which behavior is stable enough to compare.

Evaluation discipline. We split *at the user level* (80% of benign users for training; the remaining benign users plus all malicious users for test) and fit the feature scaler on training users only. A row-level split would let the model memorize a user’s baseline from training rows and trivially flag the test rows of the same user—an unrealistic UEBA evaluation.

3 Anomaly Detection Model

3.1 Architecture and training

The detector is an **LSTM autoencoder** (encoder hidden dimension 128, latent 64, 2 layers, dropout 0.2; 838,098 parameters). It consumes 30-day sliding windows of the 18-dimensional features (351,168 training sequences) and is trained *only on benign users* to reconstruct normal behavior. Anomalous sequences lie off the learned manifold and reconstruct poorly.

We minimize mean-squared reconstruction error with Adam ($\text{lr} = 10^{-3}$), gradient clipping at 1.0, and a `CosineAnnealingWarmRestarts` schedule ($T_0 = 30$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 10^{-5}$) stepped per-batch, for 50 epochs (best validation MSE 0.394). Training is tracked in MLflow (Section 11).

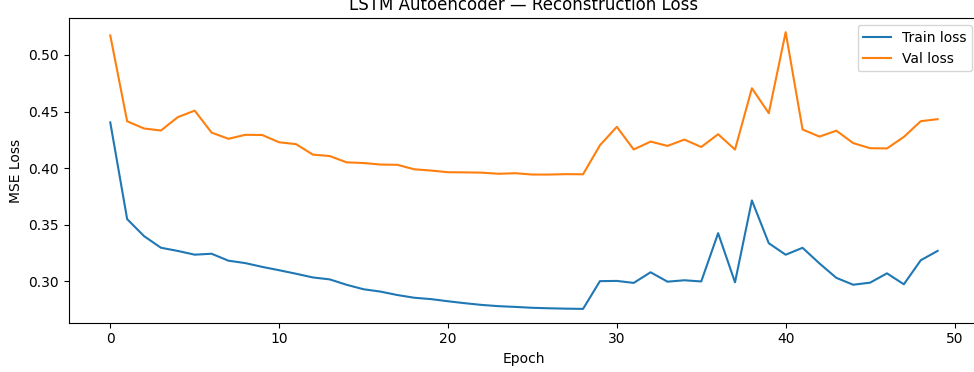


Figure 1: Training and validation reconstruction loss. The periodic bumps are the cosine warm restarts; the best checkpoint is retained.

3.2 Anomaly score

For a window x , the scalar anomaly score is the reconstruction error averaged over both time and features,

$$s(x) = \frac{1}{TF} \sum_{t=1}^T \sum_{f=1}^F (x_{tf} - \hat{x}_{tf})^2, \quad (1)$$

with $T = 30$ and $F = 18$. Per-user scores aggregate window scores by maximum, mean, top-3 mean, and 95th percentile.

4 Neyman–Pearson Calibration

A SOC has fixed analyst capacity, so the relevant constraint is a fixed false-positive rate α , not an arbitrary threshold. We calibrate the decision threshold τ on *held-out benign* scores by the empirical $(1 - \alpha)$ quantile,

$$\tau = Q_{1-\alpha}(\{s : \text{benign}\}), \quad \widehat{\Pr}(s > \tau \mid \text{benign}) \approx \alpha. \quad (2)$$

This is the Neyman–Pearson criterion: maximize detection (true-positive rate) subject to $\text{FPR} \leq \alpha$. At $\alpha = 1\%$ we obtain $\tau = 2.19$.

5 Behavioral Bifurcation Detection (BOCPD)

5.1 The model

We run BOCPD [1] on each user’s daily anomaly-score series. The core latent variable is the *run length* r_t —days since the last changepoint. BOCPD maintains the full posterior $P(r_t \mid x_{1:t})$ online. At each day the run either grows ($r_t = r_{t-1} + 1$) or resets ($r_t = 0$). With hazard $H = 1/\lambda$ (we use $\lambda = 50$) and a predictive $\pi_r = P(x_t \mid r_{t-1} = r)$, the recursion is

$$P(r_t = r+1, x_{1:t}) = P(r_{t-1} = r, x_{1:t-1}) \pi_r (1 - H), \quad (3)$$

$$P(r_t = 0, x_{1:t}) = \sum_r P(r_{t-1} = r, x_{1:t-1}) \pi_r H. \quad (4)$$

The predictive is Student- t via a Normal–Inverse–Gamma conjugate prior with unknown mean *and* variance, so the detector adapts to each user’s baseline volatility: $\pi_r = \text{St}(x_t; 2\alpha_r, \mu_r, \sqrt{\beta_r(\kappa_r + 1)/(\alpha_r \kappa_r)})$.

Table 1: Onset localization within ± 7 days by CERT scenario (BOCPD).

CERT scenario	Within ± 7 days	Character
1 (after-hours + USB + exfil)	$\approx 24\text{--}25/30$ (80%)	sharp shift
2 (job-site surfing $\rightarrow$ exfil)	1/30	gradual drift
3 (disgruntled sysadmin)	4/10	mixed

5.2 A corrected changepoint readout

A common but incorrect way to declare changepoints is to threshold $R[0, t] \equiv P(r_t = 0)$. This carries *no information*: with $S = \sum_r P(r_{t-1}=r, x_{1:t-1})\pi_r$, normalization gives

$$R[0, t] = \frac{H S}{(1 - H)S + H S} = H, \quad (5)$$

i.e. $R[0, t]$ equals the hazard at *every* step regardless of the data. The regime change instead manifests as the run-length posterior *collapsing onto short run lengths*. We therefore read off

$$c_t = \Pr(r_t \leq w) = \sum_{r=0}^w R[r, t], \quad w = 5, \quad (6)$$

and flag an onset at the first upward crossing of c_t through 0.5 after a burn-in. On a synthetic mean-shift this collapses the MAP run length from 50 to 1 at the true changepoint and fires $c_t : 0.08 \rightarrow 0.96$ with a one-day online lag.

5.3 Results

Onset-detection accuracy is strongly *scenario-dependent*, exactly as the bifurcation framing predicts (Table 1). On scenario 1 (abrupt: after-hours access + removable media + exfiltration) BOCPD localizes the onset to within ± 7 days in $\approx 80\%$ of cases, a median of 3 days, with a mean signed latency of -2 days (i.e. it tends to fire just *before* the labeled onset as behavior ramps up). On scenario 2 (gradual job-site browsing then exfiltration) it is weak: a slow drift produces no sharp run-length collapse. This is correct behavior, not a defect—a bifurcation is by definition a sharp transition.

6 Negative Results and Root-Cause Analysis

To target the gradual-drift failure we added two principled complements and evaluated them honestly (first causal alarm; $|\text{latency}| > 14$ days counts as a miss).

CUSUM and CSD fail as causal onset detectors. Used as causal detectors, both false-alarm almost immediately for every user (median latency -84 to -182 days), and scenario 2 detection is 0/30 across all transforms (raw, log, differenced). The cause is the *signal*: overlapping 30-day windows make the score series heavily autocorrelated (lag-1 autocorrelation 0.83–0.99), violating the near-independence both methods assume, and the near-constant early window yields a degenerate baseline standard deviation ($\sim 10^{-3}$) that any threshold trips. BOCPD’s sequential Bayesian model is comparatively robust to this structure. (Critical slowing down is nonetheless useful in a different role—as a pre-onset *early-warning* signal rather than a detector; Section 8.)

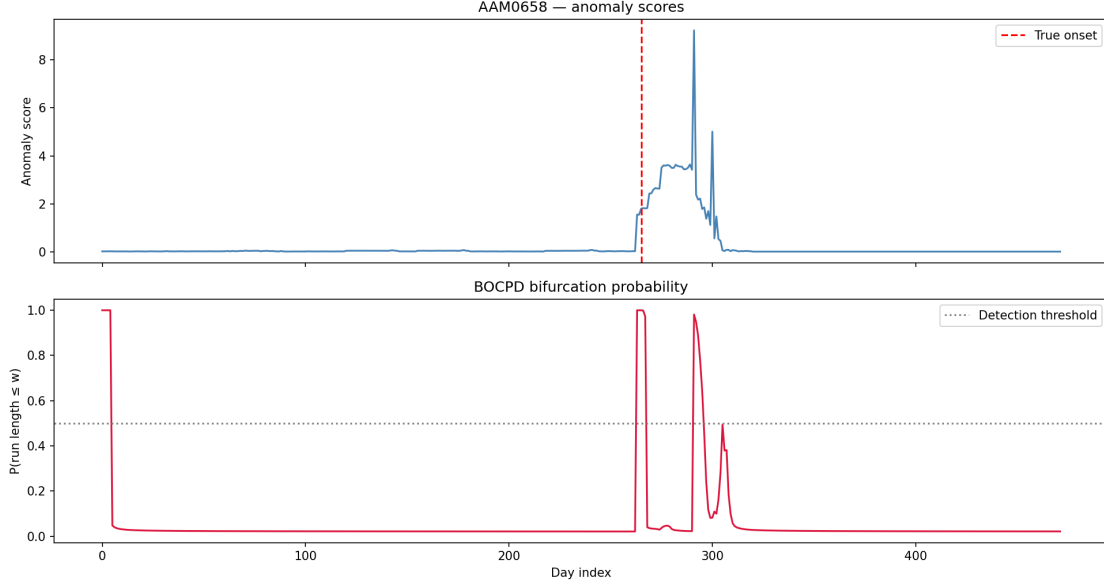


Figure 2: A scenario-1 user. Top: daily anomaly score with the labeled onset (dashed). Bottom: the run-length collapse signal c_t spiking at the regime change.

The bottleneck is the scalar score, not the detector. The scalar in Eq. (1) averages over all 18 features. At onset, the mean score rises $8.81\times$ for scenario 1 but only $1.03\times$ for scenario 2—effectively no event. Yet the underlying behavior *does* change: a single feature’s reconstruction error spikes $\sim 151\times$ (`exe_access_count`), and raw `job_site_visits` goes from near-zero to active. The averaging in Eq. (1) destroys a strong single-dimension signal. *One cannot detect what the score does not encode.*

7 Per-Feature Ablation

We therefore score per feature, averaging over time only,

$$s_f(x) = \frac{1}{T} \sum_{t=1}^T (x_{tf} - \hat{x}_{tf})^2, \quad f = 1, \dots, F, \quad (7)$$

yielding 18 error series per user, and run BOCPD per feature with a coordinated-collapse fusion (flag when ≥ 2 features collapse together, which suppresses single-feature noise). This *matches* the scalar on scenario 1 (25/30 within ± 7 days, median -3 days) and—crucially— identifies *which* behaviors drove the alert, adding explainability the scalar cannot. It confirms the dilution diagnosis quantitatively (Table 2). Pursuing scenario 2 further with sparse and offline multivariate change-point methods produced a sharper conclusion: the gradual onset is *not localizable at all*, and the reason is a property of the data, not the detector. The discriminating raw behaviors are in fact *more active before the labeled onset than after*—`job_site_visits` is active on 22.5% of pre-onset days versus 14.3% after, and `cloud_storage_visits` on 11.6% versus 4.1%—and the corresponding reconstruction error is diffusely elevated across 70–100 scattered days rather than aligned to onset. The CERT scenario-2 label marks the final exfiltration act, not a behavioral discontinuity. A bifurcation detector correctly finds bifurcations; scenario 2 has none at the label to find. (The of-

Table 2: Score change at onset (30 days before vs. after).

Scenario	Scalar score ratio	Max single-feature ratio
1 (abrupt)	$8.81\times$	$183\times$
2 (gradual)	$1.03\times$ (non-event)	$151\times$ (<code>exe_access_count</code>)

Table 3: Critical-slowng-down early-warning enrichment in the 45-day pre-onset band, scenario-1 insiders ($n = 30$).

Quantity	Value
Warning density, pre-onset band	15.6%
Warning density, baseline (elsewhere)	6.9%
Enrichment	$2.28\times$
Users with pre-onset > baseline	24/30 (chance 15/30)
Binomial test	$p = 7.2 \times 10^{-4}$
Wilcoxon signed-rank	$p = 2.1 \times 10^{-4}$

fine detector does still correctly identify `exe_access_count` as the most-changed feature for 27/30 scenario-2 users, so the per-feature view retains its explanatory value.)

8 Early Warning via Critical Slowing Down

The bifurcation framing makes a falsifiable prediction. In dynamical-systems theory, a system approaching a tipping point loses resilience, which appears as rising lag-1 autocorrelation and rising variance *before* the transition—“critical slowing down” [3]. If an abrupt insider’s onset is a genuine bifurcation, these signatures should appear in the anomaly-score series in the weeks preceding it.

We test this directly. For each scenario-1 user we compute rolling (linear-detrended) lag-1 autocorrelation and variance, and flag an early-warning day when *both* are significantly rising (Kendall’s $\tau > 0.4$ over a trailing window). We then compare the density of warning days in the 45-day pre-onset band against the same user’s baseline rate elsewhere. The effect is real and significant: warnings are $2.28\times$ **enriched** before onset (15.6% vs. 6.9% baseline), with 24/30 users showing the enrichment (chance is 15/30; binomial $p = 7.2 \times 10^{-4}$, Wilcoxon $p = 2.1 \times 10^{-4}$), and the direction of the effect is robust across detector parameters (Table 3). This is a statistically significant pre-onset signal, not an artifact.

We are careful about the claim: the 6.9% baseline rate is too high for CSD to stand alone as an alarm. Its value is as an *early-warning layer* atop BOCPD—evidence that the dynamical-systems view is not merely a metaphor but yields a measurable, predictive signature ahead of the event.

9 LLM Threat Reports

For the highest-risk users, LABAD generates analyst-ready reports. A FAISS index over the MITRE ATT&CK knowledge base retrieves techniques relevant to each user’s behavioral evidence (RAG), and a *local* LLM (Ollama, `llama3.1:8b`) writes a structured report—threat summary, severity, ATT&CK mapping, recommended actions, confidence—under an anti-hallucination instruction to use only retrieved context. Running the model locally is a deliberate design choice: behavioral logs

Table 4: Detection performance (user-level, max aggregation; 70 malicious / 256 test users).

Metric	Value
AUC-ROC	0.87
Average Precision	0.72
Malicious/benign score separation	$7.36\times$
TPR @ 10% FPR	55.7%
Neyman–Pearson @ 1% FPR (precision / recall)	70% / 43%
Precision@10 (top-3 aggregation)	100%

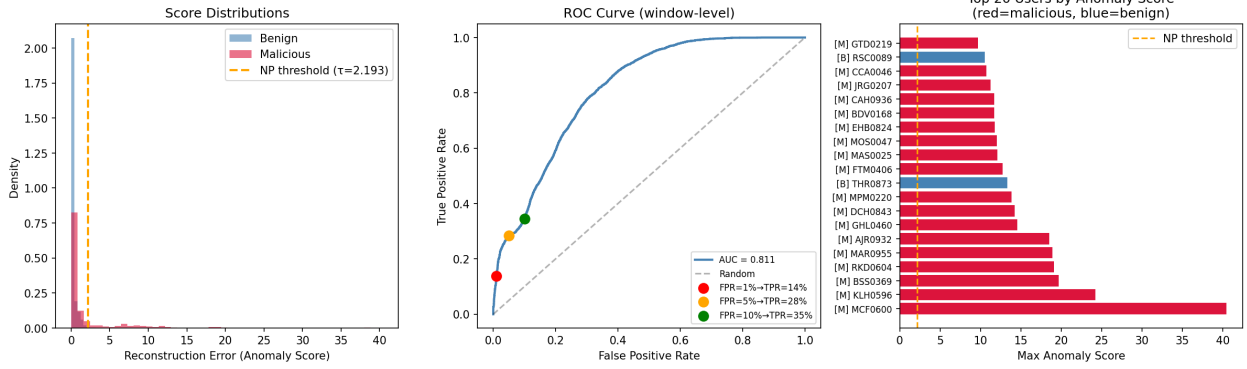


Figure 3: Score distributions (benign vs. malicious), ROC with Neyman–Pearson operating points, and the top-scored users.

are sensitive, and an air-gapped, zero-external-API pipeline is a requirement for real deployments.

10 Detection Results

Table 4 summarizes detection. The strong precision@10 means the top of the alert queue is almost entirely true insiders—the property that matters for analyst triage—while the moderate recall at 1% FPR reflects the genuine difficulty of the gradual scenarios.

11 System and Reproducibility

The pipeline is containerized with Docker and `docker-compose` (an `ollama` service plus the application). The image is CPU-only and 2.93 GB: the default PyTorch wheel bundles a multi-gigabyte CUDA stack that is dead weight in a CPU container, so we install from the PyTorch CPU index. Training runs are tracked in MLflow (parameters, per-epoch metrics, model and curve artifacts) against a SQLite backend, viewable with `mlflow ui`.

12 Limitations and Future Work

- **Gradual insiders.** Scenario 2 onset is not localizable at the labeled date because the label is not a behavioral discontinuity; a better evaluation would label the behavioral onset (start of job-site activity) rather than the final exfiltration act.

- **Early warning to alarm.** The critical-slowing-down signal is significant but not yet a standalone alarm; lowering its baseline rate (e.g. on a decorrelated or per-feature signal) could turn the early-warning enrichment into a deployable predictor.
- **Dataset.** CERT is synthetic; transfer to real telemetry is the key external-validity question.
- **Beyond human insiders.** The core capability—detect when an entity’s behavior changes regime on a stream of its actions—applies directly to monitoring autonomous AI agents for compromise, prompt-injection, or goal-drift.

13 Conclusion

LABAD shows that an insider-threat pipeline can answer *when*, not just *whether*, by detecting behavioral bifurcations with online changepoint detection. It localizes abrupt-insider onsets to a median of three days, reaches AUC 0.87 and 100% precision@10, and—through honest negative results—identifies the true bottleneck as signal representation rather than detector choice. The dynamical-systems framing is not decoration: the quantity whose collapse signals the bifurcation is literally the run-length posterior the algorithm computes.

References

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